**CANARA ENGINEERING COLLEGE**
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**
**VISEMESTER B.E. DEGREE II-IA SCHEME OF EVALUATION, OCTOBER 2019**

COMPUTER ORGANIZATION MAX 50 MARKS

| Q. No | Composition of Solution or Key Solution | Marks |
|---|---|---|
| **1.a** | USB tree structure | |
| | Diagram | 1 |
| | Labeling diagram | 1 |
| | Root Hub/ Hub | 1 |
| | Output operation of USB | |
| | Diagram | 1 |
| | Labeling diagram | 1 |
| | Explanation: Token and ACK transfer | 1+1 |
| **1.b** | Definition of synchronous bus | 1 |
| | Diagram with address and data for sender side and receiver side | 1+1+1+1 |
| | Explanation at $t_0$, $t_1$ and $t_2$ | 1+1+1 |
| **1.c** | Explanation of burst mode | 1 |
| | Functions of 8 signals | ½ x 8 = 4 |
| **2.a** | Definition | 1 |
| | Centralized arbitration | |
| | Definition | 1 |
| | Diagram | 1 |
| | Explanation | 1 |
| | Distributed arbitration | |
| | Definition | 1 |
| | Diagram | 1 |
| | Example | 1 |
| **2.b** | Input operation | |
| | Timing diagram with Address & data and handshake signal | 1+1 |
| | Explanation from t0 to t2 | 1 |
| | Explanation from t3 to t5 | 1 |
| | Output operation | |
| | Timing diagram with Address & data and handshake signal | 1+1 |
| | Explanation from t0 to t2 | 1 |
| | Explanation from t3 to t5 | 1 |
| **2.c** | Data | 1 |
| | Phase | 1 |
| | Information type | 1 |
| | Handshake | 1 |
| | Direction of transfer | 1 |
| **3.a** | 4096 x (512x8) cell array | 1 |
| | Row address latch and Row decoder | 1 |
| | Column address latch and Column decoder | 1 |
| | Sense/Write circuit with CS and R/W | 1 |
| **3.b** | Any five differences | 1+1+1+1+1 |
| **3.c** | Diagram | 1 |
| | Labeling diagram | 1 |
| | Explanation on speed | 1 |
| | Explanation on size | 1 |
| | Explanation on cost | 1 |

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| **3.d** | Direct mapping: Definition, Diagram, Example, Explanation<br>Associative mapping: Definition, Diagram, Example, Explanation<br>Set-Associative mapping: Definition, Diagram, Example, Explanation | 1+1+1+1<br>1+1+1+1<br>1+1+1+1 |
| **4.a** | Diagram:<br>Mode register with all signal<br>Refresh counter<br>Row address latch and Row decoder<br>Column address latch and Column decoder<br>Explanation<br>Definition of double data rate RAMs<br>Definition of latency<br>Definition of bandwidth | <br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 |
| **4.b** | Any five differences | 1+1+1+1+1 |
| **4.c** | Explanation of ROM<br>Explanation of PROM<br>Explanation of EPROM<br>Explanation of EEPROM<br>Explanation of Flash memory | 1<br>1<br>1<br>1<br>1 |
| **4.d** | Write buffer: Definition, , need of . write-through protocol,<br>write-back protocol<br>Prefetching: definition, , need of. perfecting through software,<br>implementation,<br>Look-up free cache: Definition, need of . issue with perfecting,<br>pipelined organization | 1+1+1<br>1<br>1+1+1<br>1<br>1+1+1<br>1 |

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|---|---|---|
| **Signature of Course Instructor with date** | **Signature of course coordinator/Expert with date** | **Head of the Department with date** |